Implementation and scientific-method review

This was one self-review pass after the core user workflow worked and the implementation was pushed and merged. It is not an independent peer review or evidence of a performance advantage. Follow-up verification is limited to the concrete defects found here and later empirical failures.

Area	Evidence examined	Finding and action
Numerical correctness	TorchCFM targets, midpoint direction/NFE, oscillator projection/normalization, deterministic forwards, finite gradients	Unit tests and fixed-batch GPU optimization passed. The readout mixes vectors before norms, preventing a constant unit-norm feature.
Learning and recovery	Full MNIST smoke workflow, exact CPU resumed/uninterrupted comparison, real GPU resume	State and EMA restoration work. Fixed-batch learning confirms nontrivial optimization. Failed-update elapsed time is now retained even when its model state is discarded.
Reliability and allocation	File locking, crash reservations, independent deadline subprocess, run status artifacts	A blocked-worker test verifies process termination. Hard-killed reservations conservatively exhaust their original per-run horizon; they cannot silently extend a fixed-time arm.
Frozen methodology	Pilot freeze, environment settings, schedule, paired RNG streams	A later `.env` edit could have changed newly launched comparison settings. Scheduled runs now load the actual saved pilot configuration through an explicit config file.
Failed evidence	Repeated command failures and report filenames	Distinct failures in one directory previously collided in generated Markdown filenames. Failure identifiers now include their attempt filenames.
Reporting completeness	Development runs alongside nested primary runs	Aggregation now recognizes the main phase under an all-runs root, producing one consolidated ledger without losing primary pair identity.
Security and publication	Positive file allowlist, symlink rejection, safetensors, tensor-only checkpoint load, credential-pattern scan	Tests exclude `.env` and unrelated files. No credential patterns were found in tracked files. Public downloaded weights match their manifest and produce exactly matching CPU predictions.
[illegible]	[illegible]	[illegible]

A CPU evaluator-cache import briefly overlapped the transient checkout performed by a PR merge with automatic local branch deletion. It failed before GPU work. The unchanged command succeeded after checkout completed. The failed attempt is retained, and active-worker merges now avoid automatic checkout transitions. The MNIST training/evaluation logs inspected during this review contained no nonfinite loss, gradient exception, or traceback.

The review did not change the backbone, optimizer defaults, data split, primary hypothesis, seed count, sampling budget, or training duration. Later empirical results must be interpreted within the protocol's limits, including architecture-specific throughput, fixed per-update EMA, small-model scope, two-seed ablations, and inference perturbations that do not alone prove a mechanism.

Sources: implementation contracts, pre-specified protocol, PyTorch serialization, SciPy exact paired permutations, NeurIPS checklist.